

KWAKU OSEI KWAKYE

Software / AI Engineer · Automation, Internal Tools & Data Dashboards

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GitHub · LinkedIn · Portfolio · [Jinsoku.ai](https://jinsoku.ai)

PROFESSIONAL SUMMARY

Software / AI Engineer with over seven years building internal tools that help operations run faster: automation applications, data dashboards, and the APIs and interfaces that support them. Most projects begin with a manual bottleneck and end with something in production that takes the repetitive work off people's hands. Recently that has meant putting LLMs and AI workflows to practical use, building them into pipelines that save real time rather than just performing well in a demo. Comfortable taking a project from initial requirements through UAT and deployment, whether working solo or with the operational teams who depend on the result. Core toolkit: Python, SQL, and modern web stacks.

CORE SKILLS

AI for Process Efficiency: LLM API integration (OpenAI, Anthropic), AI agent workflows (tool use, function calling, multi-turn), workflow automation (n8n), approval-gated / human-in-the-loop automation, NLP, speech-to-text

Automation & Backend: Python, FastAPI, Node.js / Express, Go (Gin), REST API design, SDK development, JWT authentication, microservices

Data & Dashboards: SQL (PostgreSQL, MySQL, MSSQL, Oracle, SQLite), custom web dashboards (Next.js / React + PostgreSQL), data-transformation pipelines

Languages: Python, TypeScript, JavaScript, Go, PHP, Node.js

Cloud & DevOps: AWS (EC2, Fargate, Cognito, CloudWatch), GCP, Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, CI/CD, Nginx, Prometheus

Practices: Requirements → design → UAT → deployment, Agile / Scrum, JIRA, Confluence, Git, performance optimization

PROFESSIONAL EXPERIENCE

Software / AI Engineer | Kirirom Digital, Tokyo, Japan | *Jul 2024 – Present*

- **Built AI agent orchestration and LLM-based workflows** to automate multilingual forecasting, replacing manual steps with data-driven automation.
- **Designed and shipped REST APIs (FastAPI, Node.js)** plus reusable internal service modules and SDKs, enabling product teams to integrate AI features rapidly.
- **Implemented CI/CD pipelines (GitHub Actions)**, cutting integration and release cycles by 40%.
- Deployed containerized AI workloads on AWS (EC2, Fargate, Cognito, CloudWatch) with Docker; monitored health and performance via Prometheus and CloudWatch for high availability.
- Mentored engineers on AI integration patterns and API design best practices.

Senior Backend Developer (Contract) | African Innovation Studios, New York, USA (Remote) | *Feb 2024 – Dec 2024*

- **Owned end-to-end delivery** of a scalable digital gift-card platform, from requirements and system design through build to production rollout.
- Designed high-performance REST APIs in Go (Gin) with Redis caching and PostgreSQL; implemented secure API authentication and high-throughput transaction processing.
- Managed containerized deployments (Docker, NGINX) and designed database schemas for data integrity and query performance.
- Coordinated third-party API integrations cross-team via JIRA and Confluence.

Full-Stack Software Developer | PaySwitch (FinTech), Accra, Ghana | *Mar 2023 – Jul 2024*

- **Built a Settlement Dashboard (Next.js + PostgreSQL)** that turned complex transaction data into clear operational views, translating product requirements into shipped features alongside PMs.
- **Developed an API-driven USSD device-lending platform (Go) for MTN Ghana**, automating integration across multiple payment and banking APIs.

- Designed scalable microservices for financial transaction processing under strict reliability and security requirements.
- Designed developer-friendly, documented REST APIs; implemented CI/CD workflows within cross-functional Agile teams.

Full-Stack Developer | GITPlus, Accra, Ghana | Jun 2019 – Feb 2023

- **Built internal tools and reusable backend services (Laravel, PostgreSQL)** that enabled other developers to ship features faster.
- **Delivered a national COVID-19 Vaccination Tracking System** with Redis optimization for high-traffic concurrent usage, a data-heavy operational system running at national scale.
- Led development of multi-tenant SaaS platforms: School Management (50+ institutions) and Church Management (100+ organizations).
- Architected full-stack apps (React, Next.js, PHP) with automated deployment pipelines; mentored junior developers on architecture and code quality.

SELECTED PROJECTS

- **Autonomous Content Pipeline:** End-to-end pipeline that scrapes, generates, and publishes short-form content using n8n and LLM APIs, with an approval-gated (human-in-the-loop) architecture for control over automated output. A direct example of AI applied to operational automation and efficiency.
- **agent-bridge:** Three-part system (WebSocket relay on Fly.io, desktop agent managing headless AI coding sessions, React PWA client) for remote approval and command dispatch to long-running AI agents, with human oversight designed into the automation.
- **Jinsoku.ai (Founder & Solo Developer):** Real-time AI voice translation and voice-cloning platform (42 languages, sub-300ms latency). Designed the full AI orchestration pipeline (speech-to-text, neural machine translation, voice synthesis), cloud infrastructure, and end-to-end architecture.
- **Bank File Conversion System:** Data-transformation pipeline decrypting MasterCard / Visa IPM, T112, and T11 files into readable CSV transactions (Rust, Python, PHP).

EDUCATION

BSc, Information Technology Management | University of Professional Studies, Accra (UPSA) | 2015 – 2019